

Development For Electrical Impedance Tomography (EIT) System Base on An Arduino Microcontroller

Fauzan Mala Ekaputra

Faculty of Electrical Engineering
Universiti Teknologi Malaysia
Skudai, Johor, Malaysia
mefauzan@graduate.utm.my

Nasarudin Ahmad*

Faculty of Electrical Engineering
Universiti Teknologi Malaysia
Skudai, Johor, Malaysia
*e-nasar@utm.my

Anita Ahmad

Faculty of Electrical Engineering
Universiti Teknologi Malaysia
Skudai, Johor, Malaysia
anita@utm.my

Jumadi Abdul Shukor

Faculty of Technology Engineering
Universiti Tun Hussein Onn Malaysia
Batu Pahat, Johor, Malaysia
jumadi@uthm.edu.my

Shafishuhaza Sahlan

Faculty of Electrical Engineering
Universiti Teknologi Malaysia
Skudai, Johor, Malaysia
shafis@utm.my

Mohd Zainizan Sahdan

Ittihad Maju Jaya Sdn Bhd
10-01 Jalan Waja 1, Taman Waja
Batu Pahat, Johor, Malaysia

Masturah Tunnur Mohamad Talib

Faculty of Electrical Engineering &
Technology, Universiti Malaysia Perlis
02600 Arau, Perlis, Malaysia
masturah@unimap.edu.my

Abstract—Electrical Impedance Tomography (EIT) is a non-visual imaging-based measurement method that is widely used in both industrial and biomedical fields. In EIT, there are two critical components: the current source and the data acquisition system. The fundamental principle of EIT is to measure the impedance of an object. This measurement is performed by injecting current and measuring the resulting voltage, based on Ohm's Law, for each electrode. Therefore, the current source is a crucial element in EIT. In this study, a current source is developed based on the concept of a voltage-controlled current source (VCCS). The designed current source can deliver a 50 mA amplitude and operates optimally at a frequency of 50 Hz. The current source consists of a signal generator circuit, an offset voltage regulator, an amplifier, and a Howland current source circuit. The switching module uses a 16-channel multiplexer controlled by an Arduino.

Keywords—Electrical Impedance Tomography, Howland Circuit, multiplexer switching system

I. INTRODUCTION

As industrial processes become more sophisticated, the effectiveness of sensing instruments has become increasingly important for monitoring and controlling operations. Because processes are often complex and unpredictable, there has been a long-standing demand for new sensing technologies to monitor and interpret process conditions.

Since the 1950s, tomography has emerged as a valuable instrumentation technology, capable of providing cross-sectional images of a measurement object or process [12]. Electrical tomography can serve as a powerful tool for controlling, analyzing, and optimizing various industrial processes. For example, in dredging industries, monitoring the material flow through a dredging pipe enables real-time assessment of the dredging process and control conditions. This not only improves efficiency but also helps prevent future pipeline erosion issues [12].

One key advantage of electrical-based tomography is its low cost compared to conventional tomography systems [9], [12]. Additionally, it is easy to operate and maintain [6], and

it supports high frame rates [1], making it suitable for real-time control and evaluation.

Electrical tomography (ET) can be categorized into three types: Electrical Impedance Tomography (EIT), Electrical Capacitance Tomography (ECT), and Electromagnetic Tomography (EM). This study focuses on Electrical Impedance Tomography (EIT), which utilizes an electricity-based approach.

The objective of this study is to design a low-cost and simple Electrical Impedance Tomography system using a microcontroller for educational purposes. To minimize system cost, Arduino is used to control the multiplexer, signal generator, Analog-to-Digital Converter (ADC), and to transmit boundary voltage data. Current is generated using a voltage generator and a voltage-controlled current source, then injected into 16 electrodes. These electrodes are selected for current injection or voltage collection via four multiplexers, which are automatically controlled by the Arduino. The Arduino then reads the resulting voltages and transmits the data to a PC.

II. LITERATURE REVIEW

A. Electrical Impedance Tomography

Electrical impedance tomography (EIT) is an imaging method that can estimate the electrical impedance distribution inside an object, by injecting a steady small current into the object's surface through a pair of electrodes and then calculating the impedance distribution of the object is estimated from the measured boundary voltage [2]. A simple way to understand EIT's concept is to link it with ohm's law: by injecting a current signal through a transmitting electrode pair into the conductive measuring object, the current will create voltage differences on the remaining electrode pairs [12] as shown in Fig. 1.

This procedure is only complete when all electrodes are used for injection or projection, so the cycle has as many projections as the number of electrodes, if, for instance, an

adjacent injection protocol is implemented (as exemplified in Fig. 1) [4].

Nowadays, there are many research directions for EIT technology such as the number of electrodes, reconstruction algorithms, imaging speed, noise processing, etc [11]. This study is focused on designing a simple and low-cost electrical impedance tomography system to reconstruct the image of the internal pipeline for the industrial field.

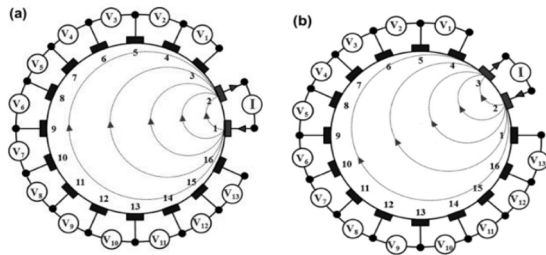


Fig. 1. EIT current injection and voltage measurement for the first (a) and second (b) projection [6]

B. EIT Image Generation

The general process of EIT imaging is to apply an incentive to the body surface of an object, form an electromagnetic field within the object, calculate the reconstruction algorithm through the obtained boundary measurement data voltages collected, and finally achieve the goal of EIT imaging [11] to generate the EIT image, some steps and some basic problems have to be solved, as shown in Fig. 2.

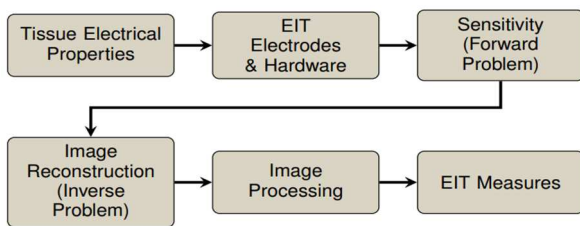


Fig. 2. Image interpretation pathway (Adler & Boyle, n.d.)

The basic EIT problem can be divided into two: In Fig. 3(a) corresponds to the forward problem, which calculates the electrical potentials in the boundary using an initial estimation of the conductivity or resistivity distribution; while (b) refers to the inverse problem that reconstructs the conductivity or resistivity distribution based on the electrical potentials measured in the boundary, through the use of a mathematical procedure such as the Gauss-Newton Method [4].

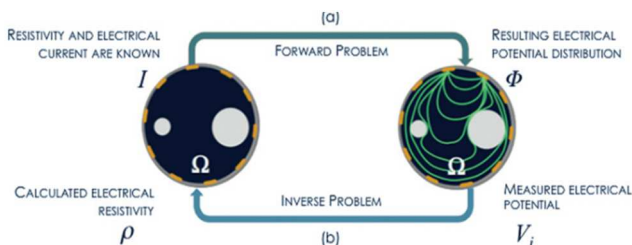


Fig. 3. Forward and inverse problems relations in the EIT [4]

C. EIT Principles

Electrical Impedance Tomography (EIT) involves two key computational problems that are essential to image reconstruction: the forward problem and the inverse problem. These problems form the mathematical foundation for interpreting voltage and conductivity distributions within a given domain.

- (1) *Forward Problem:* The forward problem of EIT is to know the conductivity distribution and boundary excitation in the field domain, solve the voltage distribution in the field domain, and get the voltage of the field boundary [11].
- (2) *Inverse Problem:* The inverse problem of EIT is to know the solved boundary voltage based on the forward problem and boundary excitation and solve the conductivity distribution in the field [11]. Or inverse problem can be understood as “inverse sensitivity” or image reconstruction (plotting)[1].

III. RESEARCH METHODOLOGY

In this study, our proposed electrical impedance tomography system will simulate by using proteus with 16 electrodes. The electrode will inject a low frequency and low amplitude AC will inject to the subject which will be represented by resistive phantom and surface potentials are acquired to reconstruct its tomographic image.

Our 16 electrodes have two main functions: first is responsible for injecting current and acquiring potentials, and second is collecting voltages responsible for reconstructing the tomographic image. Low frequency and low amplitude AC are generated by AC voltage generator and Voltage Controlled Current Source (VCCS). The two functions of the electrode for injecting current and collecting data will be switched by an automatic electrode switching module (ESM) which consists of four high-speed Complementary Metal Oxide Semiconductor (CMOS) analog multiplexers CD4067B ICs. The two analog multiplexers CD4067B ICs are acquired to inject current from VCCS to the electrodes, whereas the other two multiplexers are acquired to collect voltage potentials from the rest of adjacent pairs of electrodes.

The switching of injecting current and collecting voltage is done automatically by using multiplexors controlled by a program that is interfaced with the Arduino microcontroller.

To obtain the voltage difference between adjacent pairs of electrodes, the acquired voltages are passed through a differential amplifier, which produces a value, then is passed via a summing amplifier and finally will be read by the Arduino microcontroller. The result of one cycle of this process will be achieved 208 sets of voltage measurements.

Finally, the data fed from Arduino will be transferred to a personal computer (PC) by using Virtual Serial Port Emulator (VSPE). The complete diagram of the working system is shown in Fig. 4.

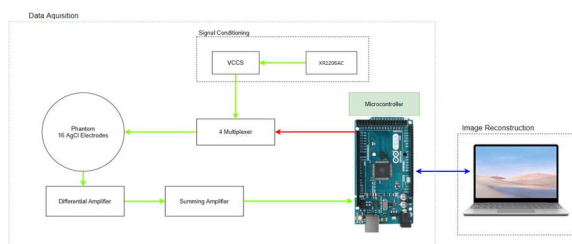


Fig. 4. Diagram of proposed Electrical Impedance Tomography

IV. HARDWARE

A. Automatic Electrode Switching Module (ESM)

The automatic Electrode Switching Module is developed by using four analog multiplexers. ICs. The first and second multiplexers are paired and responsible for injecting current from VCCS to the electrode. While the other (MUX-3 and MUX-4) are paired and switched the electrodes for collecting the voltage measurement from the rest adjacent pairs of electrodes.

The switching method of the multiplexer is using the adjacent drive method, also called as the neighboring method [6]. In order this method have good edge sensitivity and more current injection than the other method but Poor mid-mind sensitivity, measurement error, and electrode noise interference [8].

The principal of this method is current is applied through adjacent electrodes and the voltage is measured sequentially from all other adjacent electrode pairs without the pairs containing one or both the current electrodes [6] as shown in Fig. 5. For Instances, on the first cycle current injected through electrode-0 and electrode-1 and 13 voltages differences measured sequentially with pair of electrodes 2-3, 3-4, 4-5 and so on until 14-15 the switching of this electrodes is by using for loop in Arduino program. To complete all the cycle for measuring voltage the same process is done until electrode-15 and electrodes-0 injected. This technique was repeated until $13 \times 16 = 208$ sets of voltage measurements were obtained. The switching sequence of the output digital pins from the Arduino and the corresponding channel of the multiplexers for the first and second current injection are shown in Table 1.

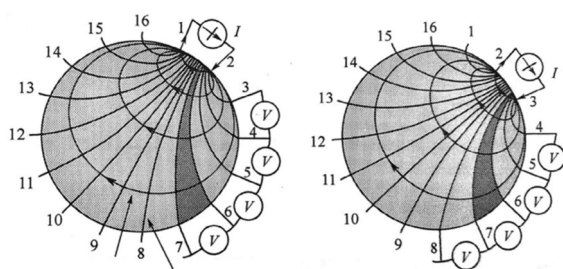


Fig. 5. Adjacent Current driven patterns (a) first cycle at left and (b) second cycle

TABLE I. THE SWITCHING SEQUENCE OF THE OUTPUT DIGITAL PINS

MUX-1				MUX-2				MUX-3				MUX-4			
D1	C1	B1	A1	D2	C2	B2	A2	D3	C1	B1	A1	D1	C1	B1	A1
0	0	0	0	0	0	0	1	0	0	1	0	0	0	1	1
								0	0	1	1	0	1	0	0
								0	1	0	0	0	1	0	1
								0	1	0	1	0	1	0	1
								0	1	1	0	0	1	1	0
								0	1	1	1	1	0	0	0
								1	0	0	0	1	0	0	1
								1	0	0	1	1	0	1	0
								1	0	1	0	1	0	1	1
								1	0	1	1	1	1	0	0
								1	1	0	0	1	1	0	1
								1	1	0	1	1	1	1	0
0	0	0	1	0	0	1	0	0	0	1	0	1	1	0	0
								0	1	0	0	0	1	0	1
								0	1	0	1	0	1	1	0
								0	1	1	0	0	1	1	1
								0	1	1	1	1	0	0	0
								1	0	0	1	1	0	0	1
								1	0	1	0	1	1	0	1
								1	0	1	1	1	0	1	1
								1	1	0	1	1	1	0	0
								1	1	0	1	1	1	0	1
								1	1	0	1	1	1	0	1
								1	1	1	1	1	1	1	1
1	1	1	1	1	0	0	0								

B. Phantom

In order to simulate the circuit for collecting data successfully work, we create Cardiff phantom circuit to simulating the body tissue [5] by shunting the resistors in the mesh can be used to introduce changes in apparent electrical conductivity at a specific area inside the phantom [10] as shown in Fig. 6.

With this phantom we attempted two experiments, the first experiments are using Cardiff phantom without load and the second experiments is with a load by shunting a circle shape in the middle of phantom as shown in Fig. 7 and 8.

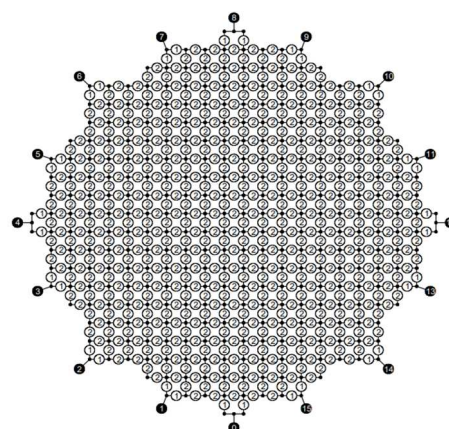


Fig. 6. Cardiff phantom [5]

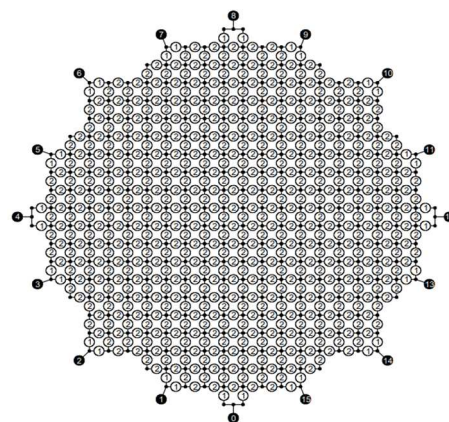


Fig. 7. Cardiff phantom without load

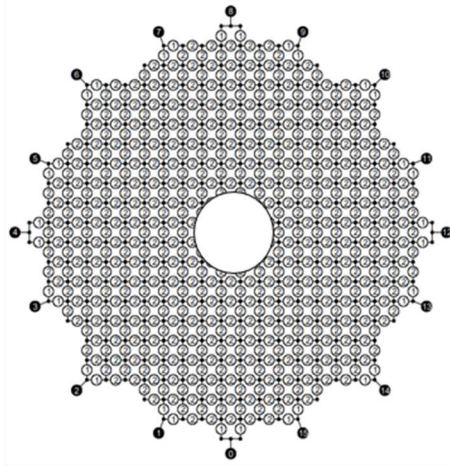


Fig. 8. Cardiff phantom with load

V. RESULTS AND DISCUSSIONS

A. Simulation

The first Experiment is conducted with the phantom subjected with no-load. The second experiment is conducted with load. The result from the simulation is shown on Table 2 and Table 3.

From the first simulation, variable x on the table indicates the electrode is injecting while the column with values shows the value of voltage potentials measurement. We can see from the Table 3. on row 1 the system is injecting due to electrode 0 and 1 and row 2 the system is injecting due to electrode 1 and 2 and so on. This also can be seen on Table 3.

From Table 2 we can see the difference value between phantom with load and no-load. This can happen due to voltage differences that occur when the load is added.

TABLE II. SIMULATION RESULT

Electrode/Cycle	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	x	x	0.006	0.012	0.009	0.01	0.008	0.006	0.006	0.007	0.009	0.008	0.009	0.012	0.041	x
2	0.025	x	x	x	0.015	0.016	0.01	0.008	0.008	0.008	0.01	0.008	0.008	0.017	0.025	0.041
3	0.006	0.046	x	x	x	0.003	0.017	0.011	0.009	0.01	0.011	0.008	0.008	0.015	0.017	0.021
4	0.012	0.025	0.046	x	x	x	0.015	0.01	0.008	0.008	0.008	0.006	0.006	0.008	0.008	0.009
5	0.009	0.015	0.046	0.025	x	x	x	0.012	0.009	0.008	0.008	0.006	0.006	0.008	0.008	0.008
6	0.01	0.016	0.033	0.046	0.016	x	x	x	0.023	0.018	0.015	0.008	0.008	0.011	0.01	0.009
7	0.008	0.01	0.017	0.015	0.025	0.044	x	x	x	0.027	0.019	0.009	0.008	0.01	0.008	0.008
8	0.006	0.008	0.011	0.01	0.012	0.019	0.04	x	x	x	0.021	0.009	0.008	0.01	0.008	0.006
9	0.006	0.008	0.009	0.008	0.008	0.023	0.044	0.035	x	x	x	0.011	0.009	0.01	0.008	0.006
10	0.007	0.008	0.01	0.008	0.008	0.016	0.037	0.04	0.025	x	x	x	0.015	0.016	0.01	0.008
11	0.009	0.01	0.011	0.008	0.008	0.015	0.019	0.021	0.036	0.046	x	x	x	0.015	0.016	0.01
12	0.008	0.008	0.008	0.006	0.006	0.006	0.009	0.009	0.011	0.025	0.046	x	x	x	0.015	0.009
13	0.009	0.008	0.008	0.006	0.006	0.006	0.008	0.008	0.009	0.015	0.046	0.025	x	x	x	0.012
14	0.021	0.017	0.015	0.008	0.008	0.011	0.01	0.01	0.01	0.016	0.033	0.046	0.016	x	x	x
15	x	0.041	0.025	0.017	0.008	0.008	0.01	0.008	0.008	0.008	0.01	0.016	0.015	0.025	x	x
16	x	x	0.04	0.021	0.008	0.008	0.009	0.008	0.006	0.006	0.008	0.01	0.009	0.012	0.016	x

TABLE III. SIMULATION RESULT II

Electrode/Cycle	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	x	x	0.015	0.011	0.008	0.01	0.008	0.006	0.007	0.011	0.01	0.007	0.008	0.02	0.04	x
2	0.025	x	x	x	0.015	0.015	0.01	0.008	0.008	0.013	0.011	0.007	0.008	0.016	0.023	0.04
3	0.005	0.046	x	x	x	0.002	0.015	0.01	0.01	0.006	0.013	0.008	0.008	0.014	0.016	0.02
4	0.011	0.025	0.046	x	x	x	0.015	0.009	0.008	0.012	0.01	0.006	0.006	0.008	0.008	0.008
5	0.006	0.015	0.046	0.025	x	x	x	0.012	0.009	0.014	0.01	0.006	0.006	0.008	0.007	0.007
6	0.01	0.015	0.032	0.046	0.015	x	x	x	0.021	0.01	0.011	0.008	0.008	0.01	0.009	0.008
7	0.006	0.01	0.015	0.015	0.023	0.038	x	x	x	0.043	0.03	0.011	0.008	0.01	0.008	0.007
8	0.006	0.008	0.01	0.009	0.012	0.015	0.057	x	x	x	0.036	0.012	0.009	0.01	0.008	0.006
9	0.007	0.008	0.01	0.008	0.009	0.021	0.038	0.061	x	x	x	0.019	0.013	0.014	0.01	0.008
10	0.011	0.013	0.016	0.012	0.014	0.03	0.043	0.057	0.038	x	x	x	0.025	0.027	0.017	0.012
11	0.01	0.011	0.013	0.01	0.01	0.01	0.021	0.03	0.046	0.042	x	x	x	0.031	0.016	0.01
12	0.007	0.007	0.008	0.006	0.006	0.009	0.011	0.012	0.019	0.037	0.044	x	x	x	0.014	0.008
13	0.008	0.008	0.008	0.006	0.006	0.008	0.008	0.009	0.013	0.025	0.042	0.013	x	x	x	0.011
14	0.02	0.016	0.014	0.008	0.008	0.01	0.01	0.01	0.014	0.027	0.031	0.044	0.015	x	x	x
15	x	0.04	0.023	0.017	0.008	0.007	0.009	0.008	0.008	0.01	0.017	0.015	0.014	0.023	x	x
16	x	x	0.04	0.02	0.008	0.007	0.008	0.007	0.006	0.008	0.012	0.01	0.008	0.011	0.015	x

B. Hardware

Based on the experiment results shown in Table 4 and Table 5, a difference value is shown between the phantom with load and without a load. Although simulation and hardware have different values, since hardware uses a water medium to create the phantom. Furthermore, the simulation does not involve conducting water, but merely testing the Arduino code and observing the data pattern.

TABLE IV. HARDWARE RESULT

Electrode/Cycle	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	x	x	0.771	0.808	0.818	0.777	0.734	0.764	0.801	0.761	0.74	0.723	0.7	0.629	0.679	x
2	0.74	x	x	x	0.766	0.777	0.749	0.729	0.736	0.763	0.737	0.722	0.709	0.689	0.633	0.66
3	0.708	0.722	x	x	0.811	0.741	0.713	0.697	0.686	0.725	0.723	0.713	0.674	0.641	0.61	0.7
4	0.718	0.711	0.723	x	x	0.813	0.776	0.716	0.713	0.717	0.735	0.701	0.657	0.637	0.606	0.666
5	0.727	0.71	0.713	0.684	x	x	x	0.785	0.735	0.731	0.713	0.709	0.703	0.649	0.632	0.704
6	0.72	0.709	0.709	0.688	0.682	x	x	x	0.733	0.726	0.714	0.701	0.685	0.641	0.603	0.709
7	0.702	0.686	0.687	0.678	0.71	0.692	x	x	0.695	0.679	0.67	0.65	0.649	0.675	0.701	0.701
8	0.707	0.707	0.708	0.717	0.724	0.709	0.675	x	x	x	0.684	0.626	0.716	0.716	0.727	0.727
9	0.715	0.695	0.707	0.726	0.713	0.688	0.654	0.717	x	x	x	0.731	0.754	0.755	0.741	0.764
10	0.688	0.688	0.712	0.715	0.699	0.685	0.676	0.723	0.721	x	x	x	0.761	0.755	0.729	0.717
11	0.671	0.691	0.692	0.681	0.654	0.624	0.682	0.709	0.709	0.7	x	x	x	0.723	0.684	0.67
12	0.703	0.715	0.71	0.691	0.654	0.663	0.727	0.713	0.713	0.742	0.676	x	x	x	0.744	0.695
13	0.674	0.685	0.685	0.664	0.654	0.656	0.693	0.682	0.701	0.68	0.695	0.663	x	x	x	0.672
14	0.673	0.682	0.693	0.659	0.629	0.663	0.701	0.702	0.695	0.691	0.66	0.68	0.689	x	x	x
15	x	0.556	0.545	0.513	0.456	0.518	0.58	0.576	0.548	0.546	0.602	0.604	0.621	0.613	x	x
16	x	x	0.683	0.572	0.568	0.58	0.576	0.657	0.573	0.585	0.551	0.609	0.579	0.547	x	x

TABLE V. HARDWARE RESULT II

Electrode/Cycle	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	x	x	0.731	0.69	0.77	0.783	0.749	0.766	0.809	0.766	0.742	0.74	0.733	0.717	0.623	0.597
2	0.702	x	x	x	0.683	0.764	0.744	0.738	0.738	0.746	0.73	0.717	0.724	0.717	0.677	0.597
3	0.697	0.71	x	x	0.616	0.739	0.731	0.688	0.683	0.712	0.722	0.723	0.709	0.687	0.63	0.63
4	0.644	0.654	0.728	x	x	x	0.613	0.783	0.728	0.708	0.709	0.737	0.72	0.71	0.686	0.617
5	0.664	0.697	0.719	0.719	x	x	x	0.612	0.732	0.733	0.683	0.697	0.732	0.722	0.707	0.617
6	0.663	0.667	0.697	0.72	0.709	x	x	x	0.777	0.725	0.706	0.696	0.761	0.723	0.704	0.609
7	0.653	0.618	0.669	0.693	0.704	0.677	x	x	x	0.706	0.691	0.677	0.66	0.678	0.68	0.609
8	0.659	0.616	0.705	0.713	0.712	0.708	0.695	x	x	x	0.716	0.732	0.719	0.723	0.72	0.608
9	0.653	0.704	0.711	0.733	0.71	0.721	0.684	0.696	x	x	x	0.714	0.749	0.729	0.715	0.714
10	0.672	0.678	0.718	0.713	0.717	0.706	0.683	0.69	0.701	x	x	x	0.747	0.732	0.706	0.7
11	0.616	0.676	0.691	0.694	0.692	0.689	0.657	0.677	0.699	0.676	x	x	x	0.695	0.672	0.649
12	0.649	0.693	0.709	0.721	0.706	0.725	0.707	0.676	0.688	0.722	0.7	x	x	x	0.727	0.692
13	0.66	0.622	0.657	0.69	0.682	0.694	0.674	0.657	0.673	0.693	0.692	0.613	x	x	x	0.682
14	0.616	0.674	0.681	0.693	0.691	0.7	0.656	0.667	0.679	0.681	0.683	0.664	0.619	x	x	x
15	x	0.442	0.519	0.568	0.548	0.522	0.551	0.629	0.584	0.604	0.543	0.564	0.482	0.476	x	x
16	x	x	1.541	1.561	1.428	1.523	1.541	1.586	1.171	1.739	1.72	0.716	0.551	1.246	1.481	x

VI. CONCLUSION

The main goal of this study was to develop hardware that could read the voltage from the phantom. There are two main parts of the circuit: a current source that must be injected in a certain order, and voltage readings that follow the same order as the current source. A current source's common current values, accuracy, adjustability, and ease of use make it a controller that automatically switches electrode voltages to obtain different impedances between electrodes. For a tomographic image to be reconstructed, all electrode data sets need to be collected. The simulation on Proteus showed that the circuit collected 208 voltage sets in 1.7 minutes with a 5.7% error rate. To calculate the error percentage, inject 0.3V 50Hz AC Sin voltage into 8 odd electrodes and 0 into the even electrodes. Furthermore, all hardware implementations are working as expected. One case, however, presents a high reading when electrode 16 injects current.

With EIDORS MATLAB, we can construct an image from all the collected data sets, and in order to make the image more accurate, we can adjust the number and diameter of electrodes. It is also possible to use another microcontroller (or another Arduino microcontroller) to speed up and improve the accuracy of data acquisition by using a microcontroller with a crystal frequency greater than 16 MHz

REFERENCES

- [1] Adler, A., & Boyle, A. (n.d.), "Electrical Impedance Tomography: Tissue Properties to Image Measures", *IEEE Transactions on Biomedical Engineering*, 64(11), 2494-2504, 2017. DOI:10.1109/TBME.2017.2728323.
- [2] Ansory, A., Prajitno, P., & Wijaya, S. K., "Design and development of electrical impedance tomography system with 32 electrodes and microcontroller", AIP Conference Proceedings, 1933, 2018. <https://doi.org/10.1063/1.5023993>
- [3] Aris, W., "Design of low-cost and high-speed portable two-dimensional electrical impedance tomography (EIT)", Article in *International Journal of Engineering and Technology*, 7(4), 6458–6463, 2018. <https://doi.org/10.14419/ijet.v7i4.23298>
- [4] Faia, P. M., Silva, R., Rasteiro, M. G., Garcia, F. A. P., Ferreira, A. R., Santos, M. J., Santos, J. B., & Coimbra, A. P., "Imaging particulate two-phase flow in liquid suspensions with electric impedance tomograph", *Particulate Science and Technology*, 30(4), 329–342, 2012. <https://doi.org/10.1080/02726351.2011.575444>
- [5] Gagnon, H., Cousineau, M., Adler, A., & Hartinger, A. E. "A Resistive Mesh Phantom for Assessing the Performance of EIT Systems", 2010.
- [6] Harikumar, R., Prabu, R., & Raghavan, S., "Electrical Impedance Tomography (EIT) and Its Medical Applications: A Review", In *International Journal of Soft Computing and Engineering (IJSCE)* (Issue 3), 2013.
- [7] Khalighi, M., Vosoughi Vahdat, B., Mortazavi, M., Hy, W., & Soleimani, M., "Practical design of low-cost instrumentation for industrial electrical impedance tomography (EIT)", 2012 IEEE I2MTC - International Instrumentation and Measurement Technology Conference, Proceedings, 1259–1263, 2012. <https://doi.org/10.1109/I2MTC.2012.6229173>
- [8] Liu, K., Wu, Y., Wang, S., Wang, H., Chen, H., Chen, B., & Yao, J., "Artificial Sensitive Skin for Robotics Based on Electrical Impedance Tomography", *Advanced Intelligent Systems*, 2(4), 1900161, 2020. <https://doi.org/10.1002/aisy.201900161>
- [9] Mansouri, S., Alharbi, Y., Haddad, F., Chabcoub, S., Alshrouf, A., & Abd-Elghany, A. A., "Electrical Impedance tomography – recent applications and developments", *Journal of Electrical Bioimpedance*, 12(1), 50–62, 2021. <https://doi.org/10.2478/joeb-2021-0007>
- [10] Mustafa Mohamedali Mohamed, M., & Elkheir Salih, M., "Design of a Simple and Low Cost Electrical Impedance Tomography System using Arduino: Proteus Simulation", للعوقة الكهربائية البسيطة والونخفض التكلفة، A project submitted in partial fulfillment for the requirements of degree of M.Sc. in biomedical engineering, 2014.
- [11] Qin, X., Ji, C., Wang, Z., & Wang, P., "Image reconstruction and simulation of fluid flow patterns in pipeline based on electrical impedance tomography algorithm", Proceedings - 2018 International Symposium on Computer, Consumer and Control, IS3C 2018, 262–265, 2019. <https://doi.org/10.1109/IS3C.2018.00073>
- [12] Wei, K., Qiu, C. H., & Primrose, K., "Super-sensing technology: Industrial applications and future challenges of electrical tomography", In *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 374(2070), Royal Society of London, 2016. <https://doi.org/10.1098/rsta.2015.0328>
- [13] Zamora-Arellano, F., López-Bonilla, O. R., García-Guerrero, E. E., Olguín-Tiznado, J. E., Inzunza-González, E., López-Mancilla, D., & Tlelo-Cuautle, E., "Development of a portable, reliable and low-cost electrical impedance tomography system using an embedded system", *Electronics (Switzerland)*, 10(1), 1–24, 2021. <https://doi.org/10.3390/electronics10010015>